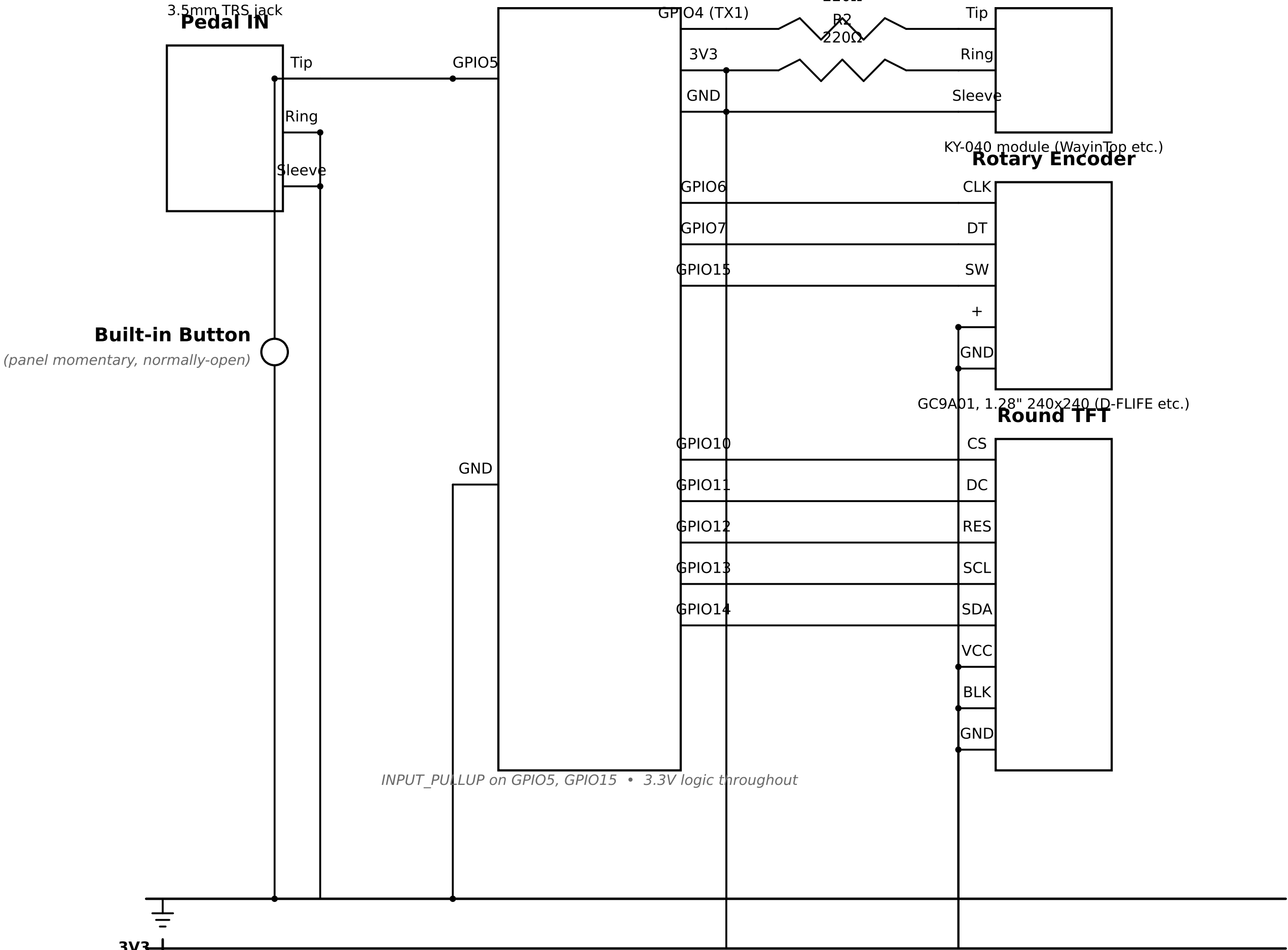


FM-1 Control Box (v2) — Schematic

ESP32-S3 Mini + TRS Pedal In + Built-in Button + TRS MIDI Out (Type A) + Rotary Encoder + 1.28" Round TFT
"Super Mini" / ESP32-S3FH4R2



Notes:

- Built-in Button is wired in parallel with the pedal jack (same GPIO5/GND nodes) -- either one alone can trigger sustain, so an external pedal is optional
- Pedal ring + sleeve are tied together so a plain mono (TS) pedal plug still grounds correctly in the TRS jack
- GPIO5 uses INPUT_PULLUP -- the button or pedal only needs to short GPIO5 to GND when pressed (normally-open momentary)
- Type A TRS MIDI: tip = signal, ring = +3V3 current source, sleeve = ground -- adapted from v1's 5V circuit, same 220Ω values; UNCONFIRMED on the bench at 3.3V (about half the MIDI-spec opto drive current) -- try 33Ω/10kΩ or add a buffer stage if the FM-1 doesn't respond
- GPIO4 (TX1) is a UART assigned in software (Serial1), not shared with the ESP32-S3's native USB-CDC -- less likely to need disconnecting before flashing than v1's Arduino was, but disconnect it if uploads ever misbehave
- If the FM-1 doesn't respond at all, it may expect Type B instead: swap Tip and Ring at the MIDI OUT jack (tip = GND, ring = signal, sleeve = GND)
- Rotary Encoder (KY-040): turning it live-auditions presets with velocity-based acceleration (MIDI Program Change per step, needs FM-1 firmware v14+); its own SW pushbutton, long-pressed, sends a full 32-voice SysEx bank dump moving the browsed preset to slot 1 ("Assign") -- see fm1_control_box.ino header comment and the project README for what Assign does and does not do (it can't press the FM-1's own A/B/C/D confirmation knob for you)
- KY-040 module has onboard pull-ups on CLK/DT/SW already -- the sketch's own INPUT_PULLUP on GPIO15 (SW) is redundant but harmless
- Round TFT: VCC and BLK (backlight) both tie straight to 3V3 -- module's listed "Driving voltage: 3-5V" makes this safe without a level shifter, but confirm your specific module's spec sheet before assuming that; backlight is always-on, no PWM dimming built yet
- Display fills with a color per preset category (piano/organ/brass/etc. -- see fm1_soundbank_data.h) so you recognize where you are while spinning through 128 presets, plus preset number/name, category name, sustain state, and an "ASSIGN SENT, turn FM-1 Knob1" prompt after a long-press